

Bulk-color v0.7 MECHANISM DEEPENING — Channel 1 + Channel 2 operator-level content beyond v0.7 outline. Per Casey continue-pulling directive. Channel 1 (g off-diagonal suppression) = Hardy-space Toeplitz operator with symbol $|z|^g$ on Shilov boundary projecting out long-extended-root contribution. Channel 2 (rank Cartan rescaling) = Cartan-basis scaling by factor rank = 2 normalizing SU(3) Cartan to A_2 standard form. Combined Π_{bulk} projector acts on $H^2(D_{IV}^5)$ via Poisson-Szegő holomorphic extension: bulk-color SU(3) emerges as the symmetry-broken extension of the boundary gauge field $V_{-}(1,0)$. Cross-link to Tier 0 v0.1.6 holographic boundary unification.

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Bulk-color v0.7 — mechanism deepening (Channel 1/2 operator content)

0. What v0.7 left open + this deepening

v0.7 absorbed Elie's two-channel decoupling FINDING (substrate $so(5) \rightarrow$ effective $A_2 = su(3)$ via $g + \text{rank}$ suppression of the $[E_{-\alpha_2}, E_{-\alpha_1+\alpha_2}] = \pm 2 \cdot E_{-\alpha_1+2\alpha_2}$ Chevalley obstruction). v0.7 said WHICH two channels ($g + \text{rank}$) drive the reduction but NOT WHAT operator does each channel's work.

This deepening fills in operator content: - Channel 1 (g off-diagonal suppression) = HARDY-SPACE TOEPLITZ OPERATOR. - Channel 2 (rank Cartan rescaling) = CARTAN BASIS SCALING. - Combined Π_{bulk} = projection onto bulk-color invariant subspace via Poisson-Szegő holomorphic extension.

Per Casey continue-pulling directive; cross-link to Tier 0 v0.1.6 holographic boundary + Lane E v0.2 expanded dictionary.

1. Channel 1 — Hardy-space Toeplitz operator (operator content)

Operator candidate:

$T_{|z|^g} :=$ Toeplitz operator on $H^2(\partial_S D_{IV}^5)$ with symbol $|z|^g$,

where $|z|$ is the radial coordinate on the Shilov boundary $\partial_S D_{IV}^5$ and $g = 7 =$ embedding dimension (substrate primary).

Why this operator: - Toeplitz operators on the Hardy space of a bounded symmetric domain are the natural multiplicative operators preserving boundary determinacy (Faraut-Korányi Ch. XIII). - The high power $|z|^g$ ($g = 7$) acts to SUPPRESS the contribution of K-types whose Bergman matrix elements are weighted toward the boundary edge. - Specifically: the long-extended root $V_{\{\alpha_1+2\alpha_2\}}$ of B_2 lives at the highest weight in the K-type tower; its Bergman matrix elements are concentrated at $|z| \rightarrow 1$ (boundary edge); the Toeplitz operator $T_{\{|z|^g\}}$ suppresses this contribution via the high-power decay near the edge. - After suppression, the effective algebra on $H^2(\partial_S D_{IV}^5)$ reduces from B_2 to A_2 (the remaining root structure).

Substrate-primary content: $g = 7 = \text{embedding dimension} = p+q$ of $SO(5,2)$ signature. The Toeplitz symbol $|z|^g$ uses substrate-primary g as its exponent, not arbitrary. This is substrate-natural.

Falsifier (F-bulk-1): if $T_{\{|z|^k\}}$ with $k \neq g$ (e.g., $k = 5$ or 6) gives the same effective A_2 reduction, then g is NOT load-bearing and the Channel-1 mechanism is misidentified. Specific test: Elie numerical computation of effective Chevalley structure constants after Toeplitz suppression at varying k ; only $k = g = 7$ should give 3/3 positive-positive match.

2. Channel 2 — Cartan rescaling operator (operator content)

Operator candidate:

$R_{\text{Cartan}} := \text{scaling operator on K-Cartan subspace by factor rank} = 2.$

Specifically: the substrate's $K = SO(5) \times SO(2)$ has 2-dimensional Cartan subalgebra (rank 2). $SU(3) = A_2$ also has 2-dimensional Cartan. The K-Cartan generators (H_1, H_2) need to be rescaled to match $SU(3)$'s standard Cartan basis. The rescaling factor is $\text{rank} = 2$.

Why this operator: - The K-Cartan basis from $SO(5)$ is normalized to Killing form of B_2 ; $SU(3)$ Cartan basis is normalized to Killing form of A_2 . - The two normalizations differ by the ratio of the long-root-length² / short-root-length² of B_2 , which is 2 (B_2 has roots of two lengths, ratio $\sqrt{2}$; lengths² ratio = 2 = rank). - So the Cartan rescaling by factor 2 = rank converts B_2 -normalized Cartan to A_2 -normalized.

Substrate-primary content: $\text{rank} = 2$ is THE substrate's defining primary. The rescaling factor IS rank, not arbitrary. Substrate-natural identification.

Falsifier (F-bulk-2): if Cartan rescaling by factor $\neq \text{rank}$ still produces effective $A_2 = \mathfrak{su}(3)$ closure, then rank is NOT load-bearing. Specific test: Elie verify that ONLY $\text{rank} = 2$ rescaling gives $\mathfrak{su}(3)$ -standard Cartan-Weyl commutation relations.

3. Combined Π_{bulk} projector via Poisson-Szegő extension

Per Tier 0 v0.1.6 boundary unification: the substrate's REAL degrees of freedom live on the Shilov boundary; the interior bulk $H^2(D_{IV}^5)$ is the holomorphic extension via Poisson-Szegő.

The bulk-color projection Π_{bulk} acts on $H^2(D_{IV}^5)$ via:

$$\Pi_{\text{bulk}} = P_{\text{Szegő}} \circ (R_{\text{Cartan}} \circ T_{\{|z|^g\}}) \circ P_{\text{Szegő}}^{-1},$$

where $P_{\text{Szegő}}$ is the Poisson-Szegő boundary-to-interior map and $R_{\text{Cartan}} \circ T_{\{|z|^g\}}$ acts on the boundary side via Channels 1+2.

Reading: bulk-color $SU(3)$ emergence happens BOUNDARY-SIDE (Channels 1+2 act on $\partial_S D_{IV}^5$); the interior bulk inherits the $SU(3)$ structure via the Poisson-Szegő extension. This is consistent with Tier 0 v0.1.6's "bulk is holomorphic extension of boundary."

Net result: the boundary substrate field theory (with full $\mathfrak{so}(5) \times \mathfrak{so}(2)$ structure) gets PROJECTED to bulk-color invariant subspace; the interior holomorphic extension of this projection carries effective $SU(3) = \text{bulk-color}$. QCD-like gauge structure emerges via the boundary-to-bulk projection mechanism.

4. Cross-link to Tier 0 v0.1.6 + Lane E v0.2

Tier 0 v0.1.6 boundary unification: Π_{bulk} lives on the boundary; bulk $SU(3)$ is the extension. This makes bulk-color a BOUNDARY mechanism with bulk manifestation — consistent with the substrate's holographic-principle structure.

Lane E v0.2 candidates: - Gluons (candidate 7): $V_{\text{bulk}}(1, 1)$ bulk-projected via Π_{bulk} . The 8 gluons live in the bulk after Π_{bulk} projection; their Toeplitz structure $8 = 3 T_a + 3 T_a^\dagger + 2 K$ -Cartan is the action of Channels 1+2 on $V_{\text{bulk}}(1, 1)$. - Up quark, down quark (candidates 5, 6): $V_{\text{bulk}}(1, 0)$ and $V_{\text{bulk}}(0, 1)$ bulk-projected; carry effective $SU(3)$ color charge (triplet $\oplus$ anti-triplet $\oplus$ singlet) via Π_{bulk} . - W boson charged (candidate 3): lives in bulk-Shilov mixed region; partial Π_{bulk} projection (charge component bulk-localized; gauge-invariant Cartan Shilov-anchored).

L4 v0.2 mass framework: the L4 v0.2 Bergman matrix element computation for T190 $(24/\pi^2)^6$ lives on the Shilov-side via the spinor radial tower $V_{\text{bulk}}(1/2, 1/2)^{\wedge\{n\}}$. Quarks (bulk-color) need a DIFFERENT mass formula since they're bulk-localized post- Π_{bulk} ; the bulk-mass formula = Bergman matrix elements on $V_{\text{bulk}}\lambda$ -bulk-projected, multi-week target.

5. Calibration #35-candidate audit revisited

v0.7's flag on $g + \text{rank} = N_c^2$ appearing in two contexts (Weinberg + bulk-color) gets sharpened by this deepening:

Channel 1 uses g as Toeplitz exponent (operator-level content). Channel 2 uses rank as Cartan rescaling factor (operator-level content). Both are operator-distinct uses of the substrate-primary integers; they are NOT the same operator's action in two contexts.

Compared to Weinberg's $\text{rank}/N_c^2 = 2/9$ (which uses rank and N_c arithmetically, not as operators): - Weinberg: arithmetic ratio of substrate primaries in EW partition. - Bulk-color Channel 1: Toeplitz operator with exponent g . - Bulk-color Channel 2: Cartan rescaling by factor rank .

Per Calibration #35-candidate: the SHARED substrate-primary IDENTITIES (rank, g, N_c) are NOT independent uses; they're appearances of the same substrate primaries in multiple mechanisms. But the OPERATORS that act in each context (arithmetic partition vs Toeplitz vs Cartan rescaling) ARE distinct mechanisms.

Honest reading: substrate primaries are SHARED; mechanisms are DISTINCT; null-model independence accounting per Calibration #35-candidate applies to mechanism count, not primary count. (Cal cold-read #188 will verify this disposition.)

6. Verification target (Elie multi-week)

For Bulk-color v0.7 mechanism content to elevate from CANDIDATE to RIGOROUS:

Step 1: Numerical computation of $T_{\text{bulk}}\{|z|^\wedge g\}$ action on $V_{\text{bulk}}(1,1)$ (long-extended root contribution); verify 3/3 positive-positive Chevalley match after suppression. ELIE Toy 3661+ target.

Step 2: Cartan rescaling by $\text{rank} = 2$ verified to give $SU(3)$ -standard normalization. ELIE Toy 3661+ continued.

Step 3: Combined Π_{bulk} action on $V_{\text{bulk}}(1, 1) \rightarrow 8$ -fold $\mathfrak{su}(3)$ Cartan-Weyl basis (Lane E gluons candidate verification). ELIE multi-week.

Step 4: Falsifier tests (F-bulk-1 + F-bulk-2): vary $k \neq g$ and rescaling factor $\neq \text{rank}$; confirm only $g=7 + \text{rank}=2$ give $SU(3)$ closure. ELIE multi-week.

Promotion: CANDIDATE $\rightarrow$ RIGOROUS via Steps 1-4 + Cal #188 cold-read PASS.

7. Honest scope + tier

RIGOROUS (this deepening): - Toeplitz operators T_f on $H^2(\partial_S D_{IV}^5)$ are well-defined (FK Ch. XIII). - Cartan rescaling by factor 2 between B_2 and A_2 Cartan normalizations is standard rep theory (Humphreys 1972). - Poisson-Szegő extension is the unique holomorphic extension from boundary (Hardy decomposition).

CANDIDATE (this deepening's load-bearing): - Channel 1 = $T_{\{|z|^g\}}$ with $g = 7 =$ embedding dimension. - Channel 2 = R_{Cartan} by factor $\text{rank} = 2$. - Combined $\Pi_{\text{bulk}} = P_{\text{Szegő}} \circ (R_{\text{Cartan}} \circ T_{\{|z|^g\}}) \circ P_{\text{Szegő}}^{-1}$ projection. - Bulk-color $SU(3)$ emergence = holomorphic extension of boundary-projected effective A_2 .

FRAMEWORK (Elie multi-week): - Numerical verification of $T_{\{|z|^g\}}$ suppression of long-extended root. - Falsifier tests F-bulk-1 + F-bulk-2 (confirm $g=7 + \text{rank}=2$ are load-bearing, not arbitrary). - Π_{bulk} action on $V_{\{1, 1\}}$ yielding 8-fold gluon basis. - Cross-anchor with Lane E gluons + quarks bulk-color candidates.

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: this deepening provides operator-level content for v0.7 mechanism (not just arithmetic identity); per Calibration #35-candidate audit, shared substrate primaries (g, rank) used in operator-distinct ways across contexts $\rightarrow$ mechanisms are distinct even if primaries shared.

8. Routing

$\rightarrow$ Casey: bulk-color v0.7 mechanism content deepened per pull-continuously directive. Channel 1 = Hardy Toeplitz $T_{\{|z|^g\}}$; Channel 2 = Cartan rescaling by rank. Combined Π_{bulk} = boundary-to-bulk projection via Poisson-Szegő. Operator-level statement; multi-week verification via Elie Toy 3661+.

$\rightarrow$ Elie: explicit operator candidates for Channels 1+2. Verification targets: F-bulk-1 (only $g=7$ gives 3/3 Chevalley match), F-bulk-2 (only $\text{rank}=2$ gives $SU(3)$ Cartan-standard); Steps 1-4 multi-week. Coordinate with Toy 3660 Mehler kernel work for bulk-color matrix elements.

$\rightarrow$ Keeper: K-audit pre-stage K-audit-Bulk-Color-v0.7-deepening; Session 2 absorb into Tier 0 v0.2 connection to Lane E gluons + quarks bulk-localization.

$\rightarrow$ Grace: catalog operator content ($T_{\{|z|^g\}}, R_{\text{Cartan}}, \Pi_{\text{bulk}}$) at CANDIDATE; cross-reference to Tier 0 v0.1.6 + Lane E v0.2 + Calibration #35-candidate accounting.

$\rightarrow$ Cal: cold-read #188 specific concerns: (1) shared substrate primaries vs distinct operators framing (does the Calibration #35-candidate audit treat OPERATOR-distinct uses of shared primaries as independent or not?); (2) Toeplitz symbol $|z|^g$ exponent choice (is g forced or arbitrary?); (3) Π_{bulk} projection mechanism vs post-hoc construction.

$\rightarrow$ me: continuing pulls per Casey directive — next: L4 v0.2 multi-week extension OR P1 v0.8 pre-revision notes.

— Lyra, Bulk-color v0.7 MECHANISM DEEPENING (Channels 1+2 operator content). Channel 1 = Hardy Toeplitz $T_{\{|z|^g\}}$ ($g = 7 =$ embedding dimension); Channel 2 = Cartan rescaling by $\text{rank} = 2$; Combined $\Pi_{\text{bulk}} = P_{\text{Szegő}} \circ (R_{\text{Cartan}} \circ T_{\{|z|^g\}}) \circ P_{\text{Szegő}}^{-1}$ projection. Bulk-color $SU(3)$ emergence = boundary-projected A_2 + holomorphic extension. Cross-anchored to Tier 0 v0.1.6 boundary unification + Lane E v0.2 gluons/quarks bulk-localization. Falsifiers F-bulk-1 + F-bulk-2 explicit; ELIE Toy 3661+ multi-week verification.